ECA08 – ANALOG AND DIGITAL COMMUNICATION

CO2 Engineering Case Study Report

FM Broadcast System: From Angle-Modulation Selection to Noise-Performance Deployment

Item	Details
Name	U. Guru Prasad Reddy
Roll No.	192312683L
Course / Section	ECA08 – Analog and Digital Communication
Course Outcome	CO2 – Angle modulation, bandwidth, noise, pre-/de-emphasis
Process followed	REQUIREMENT → ALTERNATIVES → ANALYSIS → SELECTION → DESIGN → SIMULATION → VERIFICATION → DEPLOYMENT DECISION

Final engineering recommendation in one line

Deploy: YES. Wideband FM, carrier 100 MHz, f_m = 15 kHz, peak deviation 75 kHz (β = 5), Carson bandwidth 180 kHz inside a 200 kHz channel, 75 µs pre-emphasis/de-emphasis, superheterodyne receiver (10.7 MHz IF, 200 kHz IF filter, limiter + quadrature/PLL discriminator), operating IF C/N ≈ 15–20 dB (≥ 5 dB above the ≈ 10 dB threshold).

Note on tools. MATLAB was not available in the environment used to prepare this report. All simulation figures and numbers were generated by an algorithm-for-algorithm equivalent in Python (NumPy/SciPy); the corresponding MATLAB script is given in Appendix A. Run it in MATLAB and replace the figures before submission if your instructor requires MATLAB-generated plots. The MATLAB script itself has not been executed here.

Case-Study Problem Statement and Assumptions

A regional FM broadcast station upgrades its transmitter and receiver chain. The design must deliver acceptable recovered-audio quality within practical spectrum, power and complexity limits. Given data and the parameters this report must determine are summarised below.

Parameter	Given	To be decided (this report)
Information signal	Broadcast audio, f_m(max) = 15 kHz	Audio processing, pre-emphasis
Carrier	100 MHz (within 88–108 MHz band)	Transmitter architecture
Peak deviation	75 kHz (initial)	Final deviation, β, bandwidth
Channel	AWGN + practical receiver/interference limits	Operating C/N, margin
Receiver	Limiter + frequency discriminator	IF, selectivity, detector type
Reference systems	AM, DSB-SC, FM, PM	Noise benchmark
Platform	MATLAB	Verification set

Assumptions

- Mono audio service (stereo multiplex and RDS are discussed only as extensions).
- Channel noise is additive, white, Gaussian with two-sided PSD $N_0/2$; the signal is received with a flat (distortion-free) frequency response.
- Receiver noise figure 5 dB, antenna noise temperature 290 K, ideal brick-wall IF filter of 200 kHz in simulation.
- Channel raster 200 kHz (as used in the 88–108 MHz band under US practice); the same design works on a 100 kHz raster only with reduced deviation.
- Sinusoidal tests use a single tone; program-material statistics are discussed separately where they change the answer.
- Pre-emphasis time constant $\tau = 75$ μs ($f_1 = 1/2\pi\tau = 2.12$ kHz); 50 μs is the alternative used in Europe.

Stage 1 – Interpret the Broadcast System Requirement (10 marks)

1.1 Communication-system block diagram

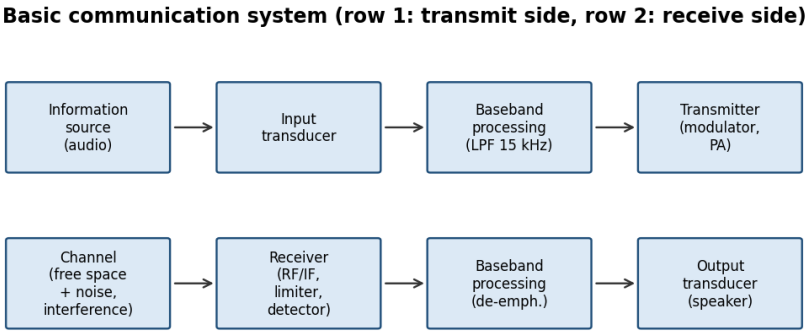


Figure 1: Basic communication system.

The **source** produces the audio message m(t); the **input transducer** (microphone/studio feed) converts it to an electrical baseband signal; **baseband processing** limits it to 15 kHz and applies pre-emphasis; the **transmitter** modulates the 100 MHz carrier and amplifies it; the **channel** (free space) attenuates the signal and adds thermal noise and interference; the **receiver** selects the station, amplifies, limits and demodulates; de-emphasis and the loudspeaker complete the chain.

1.2 Need for modulation; limits of direct baseband transmission

- **Antenna size:** an efficient antenna is of the order of $\lambda/4$. At 15 kHz, $\lambda = 20$ km ($\lambda/4 = 5$ km); at 100 MHz, $\lambda = 3$ m ($\lambda/4 = 75$ cm).
- **Multiplexing:** all stations occupy the same audio band 20 Hz–15 kHz; baseband transmission would make them overlap. Modulation shifts each to its own carrier (frequency-division multiplexing).
- **Propagation:** VHF carriers give line-of-sight coverage; audio frequencies do not radiate usefully.
- **Noise/interference trade:** wideband modulation (FM) exchanges bandwidth for SNR – impossible at baseband.
- **Hardware:** at VHF, filters, amplifiers and antennas are small and practical.

1.3 Angle modulation; FM versus PM

In angle modulation the carrier amplitude is constant and the message is carried in the instantaneous phase $\theta_i(t)$. Writing $s(t) = A_c \cos \theta_i(t)$, the instantaneous frequency is $f_i = (1/2\pi)\, d\theta_i/dt$.

	FM	PM
Phase	$\theta_i = 2\pi f_c t + 2\pi k_f \int m(\tau)d\tau$	$\theta_i = 2\pi f_c t + k_p m(t)$
Inst. frequency	$f_i = f_c + k_f m(t)$	$f_i = f_c + (k_p/2\pi)\, dm/dt$
Deviation depends on	message amplitude only	message amplitude and frequency
Relationship	PM of $\int m(t)$ = FM of m(t)	FM of dm/dt = PM of m(t)

1.4 Message bandwidth and carrier requirements

$W = 15$ kHz (high-fidelity audio). $f_c = 100$ MHz lies in the 87.5/88–108 MHz VHF Band II. $f_c/W \approx 6667$, so the narrow-band-carrier assumption ($f_c \gg$ bandwidth) holds very well and the complex-envelope model used in Stage 7 is exact for practical purposes.

1.5 Constant-average-power operation

For $s(t) = A_c \cos[\theta_i(t)]$ the instantaneous amplitude is always A_c, so the average power is $P = A_c^2/2$ **regardless of m(t)**, modulation index or deviation. Equivalently, for tone modulation the Bessel-series amplitudes satisfy $\Sigma J_n^2(\beta) = 1$. Consequences: (i) the transmitter can use efficient Class-C/E power amplifiers; (ii) increasing deviation does not change transmitted power – it only redistributes power from the carrier into sidebands ($J_0(5) = -0.178$, i.e. only 3.2 % of the power remains in the carrier); (iii) the bandwidth-for-SNR trade of FM costs no extra power.

1.6 Noise sources, Gaussian and white characteristics

Source	Nature	Effect
Thermal noise (receiver front end)	White, Gaussian, $N_0 = kT$	Dominant AWGN; sets sensitivity
Antenna / sky / man-made noise	Broadband, approx. Gaussian, impulsive in urban areas	Raises effective noise temperature
Co-/adjacent-channel interference	Deterministic or FM-like	Capture effect; need protection ratio
Multipath fading, image and spurious responses	Signal-dependent	Distortion, noise bursts
Transmitter/oscillator phase noise	Close-in noise	Limits low-level SNR

Thermal noise results from the random motion of a very large number of electrons, so by the central limit theorem it is Gaussian with zero mean; its PSD is flat (white) up to far above the VHF band, $S_N(f) = N_0/2$ with $N_0 = kT$. After a band-pass filter of bandwidth B it can be written $n(t) = n_I(t)\cos 2\pi f_c t - n_Q(t)\sin 2\pi f_c t$ where n_I and n_Q are independent, zero-mean Gaussian processes, each with the same power as n(t) itself (N_0B for a one-sided band of B, i.e. $2N_0W$ for a DSB signal). For the 200 kHz IF: $N = kTB = -174$ dBm/Hz $+ 10\log(2\times10^5) =$ **-121 dBm**; with 5 dB noise figure the floor is **-116 dBm**.

1.7 Practical constraints

- Spectrum: 200 kHz channel; adjacent-channel leakage must stay within the regulatory emission mask.
- Power: constant-envelope PA, EIRP limited by licence class; coverage set by required C/N (≈ 15 dB) at the contour.
- Receiver: low-cost superheterodyne; limiter + discriminator; image and adjacent-channel rejection.
- Audio quality: 30 Hz–15 kHz, ≥ 40–50 dB SNR, low distortion (< 1 %); pre-emphasis must not cause over-deviation.

Stage 1 Deliverable – Broadcast Communication Design Requirement Statement

Item	Requirement
Message	Audio, 15 kHz maximum frequency
Carrier	100 MHz (FM Band II)
Initial deviation	$\Delta f = 75$ kHz → $\beta = \Delta f/W = 5$
Channel	AWGN ($N_0 = kT$, NF 5 dB) + interference; −121 dBm noise in 200 kHz
Receiver	Superheterodyne, limiter + frequency discriminator, C/N ≥ 15 dB at contour
Bandwidth limit	Fit in 200 kHz channel (Carson ≈ 180 kHz)
Audio quality	Recovered SNR ≥ 45 dB mono, flat 30 Hz–15 kHz after de-emphasis
Implementation	Constant-envelope PA, stable carrier (±2 kHz), low-cost receiver

Case Study Decision 1

I recommend a **wideband constant-envelope angle-modulation system (f_c = 100 MHz, W = 15 kHz, Δf = 75 kHz, β = 5, 200 kHz channel, operating C/N ≈ 15 dB)** as the system requirement/operating basis because baseband transmission is impractical, the constant average power of angle modulation allows an efficient transmitter, and the 200 kHz spectrum allocation is large enough to trade bandwidth for a much larger SNR than any linear modulation can give at the same transmitted power.

Stage 2 – Select the Angle-Modulation Scheme (15 marks)

2.1 Mathematical expressions (sinusoidal message)

Let m(t) = A_m cos(2πf_m t).

$$FM:\ s(t) = A_c \cos[2\pi f_c t + \beta \sin(2\pi f_m t)],\quad \beta = \Delta f / f_m = k_f A_m / f_m$$

$$PM:\ s(t) = A_c \cos[2\pi f_c t + k_p A_m \cos(2\pi f_m t)],\quad \beta_p = k_p A_m$$

Instantaneous phase and frequency (FM tone): $\theta_i(t) = 2\pi f_c t + \beta \sin 2\pi f_m t$; $f_i(t) = f_c + \Delta f \cos 2\pi f_m t$, so f varies between f_c ± Δf. **PM tone:** $\theta_i = 2\pi f_c t + \beta_p \cos 2\pi f_m t$; $f_i = f_c - \beta_p f_m \sin 2\pi f_m t$, so the peak deviation is Δf = β_p f_m and **grows with message frequency**.

2.2 Modulation index and classification

Case	Δf	f_m	β	Class
Design point (top audio frequency)	75 kHz	15 kHz	5.0	Wideband FM
Mid audio	75 kHz	5 kHz	15	Wideband FM
1 kHz test tone	75 kHz	1 kHz	75	Wideband FM
Narrow-band reference	75 Hz	15 kHz	0.005	NBFM (β ≪ 1)

Narrow-band operation requires β ≲ 0.3 (peak phase error ≪ 1 rad); the design β = 5 ≫ 1, so FM is **wideband**: many significant sidebands and an SNR gain proportional to β². The quantity 5 is the "deviation ratio" D = Δf_max/W of broadcast FM.

2.3 Effect of deviation and message frequency on bandwidth

Carson's rule: B_T ≈ 2(Δf + f_m) = 2f_m(β + 1). For FM, Δf is fixed by amplitude, so B_T changes only slightly with f_m (from 2Δf for f_m → 0 up to 2(Δf+W)). For PM, Δf = β_p f_m, so B_T = 2f_m(β_p + 1) is **proportional to f_m**.

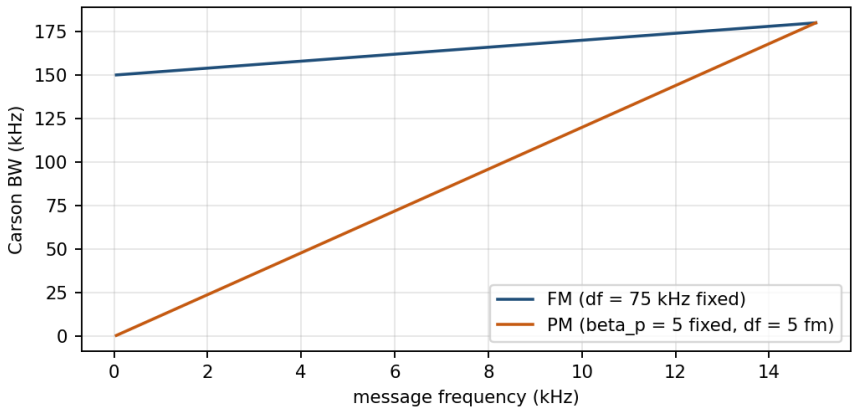


Figure 2: Carson bandwidth versus message frequency – FM (Δf = 75 kHz) and PM (β_p = 5).

2.4 Quantitative FM–PM comparison

Criterion	FM (Δf = 75 kHz)	PM (β_p = 5, same peak deviation at 15 kHz)
Modulation index	β = Δf/f_m: varies 5 … 1500 with f_m	β_p = 5 constant
Peak deviation vs f_m	Constant 75 kHz	Δf = 5f_m: 75 kHz at 15 kHz, 5 kHz at 1 kHz
Bandwidth (Carson)	2(75+f_m): 180 kHz at 15 kHz, 152 kHz at 1 kHz	2·6·f_m: 180 kHz at 15 kHz, only 12 kHz at 1 kHz
Average power	A_c²/2 (constant)	A_c²/2 (constant)

Criterion	FM ($\Delta f = 75$ kHz)	PM ($\beta_p = 5$, same peak deviation at 15 kHz)
Output SNR (tone, above threshold)	$1.5(\Delta f/W)^2\gamma = 37.5\gamma$ (+15.7 dB)	$0.5\beta_p^2\gamma = 12.5\gamma$ (+11.0 dB)
Noise PSD at detector	$\propto f^2$ (parabolic) – tamed by de-emphasis	White (flat) – no benefit from de-emphasis trick
Generation	VCO (direct) or integrator + phase modulator (Armstrong)	Phase modulator (simple, crystal-stable)
Detection	Limiter + discriminator / PLL	Discriminator followed by integrator
Complexity	Moderate; mature broadcast hardware	Moderate; non-standard for broadcast
Typical use	Analog broadcast, VHF audio	Digital phase modulation (PSK), some data links

For equal peak frequency deviation at the highest audio frequency, FM gives **3× (4.77 dB) better output SNR** than PM, because the PM output is proportional to the phase, so the low-frequency signal components carry less deviation while the noise is white; FM, by contrast, has most of its noise at high frequencies where pre-/de-emphasis can remove it. In addition, PM would use much less bandwidth for low-frequency audio, i.e. it would waste the allocated 200 kHz channel and not exploit it to improve SNR.

Case Study Decision 2

I recommend **frequency modulation (wideband FM, $\beta = 5$, $\Delta f = 75$ kHz)** because it gives +15.7 dB (tone) over baseband versus +11.0 dB for PM at the same top-frequency deviation, uses the whole 200 kHz channel productively, is detected by simple limiter–discriminator receivers, and its parabolic noise spectrum allows a further ≈ 13 dB (ideal) improvement through pre-/de-emphasis. PM was not selected because its SNR is 4.77 dB lower, its occupied bandwidth varies strongly with audio frequency (wasting spectrum at low frequencies), and it is not compatible with the established FM receiver base.

Stage 3 – FM Spectrum and Transmission Bandwidth (15 marks)

3.1 Bessel-function spectrum

Expanding exp[jβ sin 2πf_m t] in a Fourier series gives

$$s(t) = A_c \Sigma_n J_n(\beta) \cos[2\pi(f_c + n f_m)t], \quad n = -\infty \ldots \infty$$

so the spectrum is a carrier (J_0) plus sidebands at f_c ± n f_m with amplitudes $|J_n(\beta)|$; $J_{-n} = (-1)^n J_n$. For β = 5, f_m = 15 kHz, A_c = 1:

n	Offset n·f_m	$J_n(5)$	Level	Cumulative power in ±n pairs
0	0 kHz	-0.1776	-15.0 dB	3.15 %
1	15 kHz	-0.3276	-9.7 dB	24.62 %
2	30 kHz	0.0466	-26.6 dB	25.05 %
3	45 kHz	0.3648	-8.8 dB	51.67 %
4	60 kHz	0.3912	-8.2 dB	82.28 %
5	75 kHz	0.2611	-11.7 dB	95.92 %
6	90 kHz	0.1310	-17.7 dB	99.36 %
7	105 kHz	0.0534	-25.5 dB	99.93 %
8	120 kHz	0.0184	-34.7 dB	99.99 %
9	135 kHz	0.0055	-45.2 dB	100.00 %

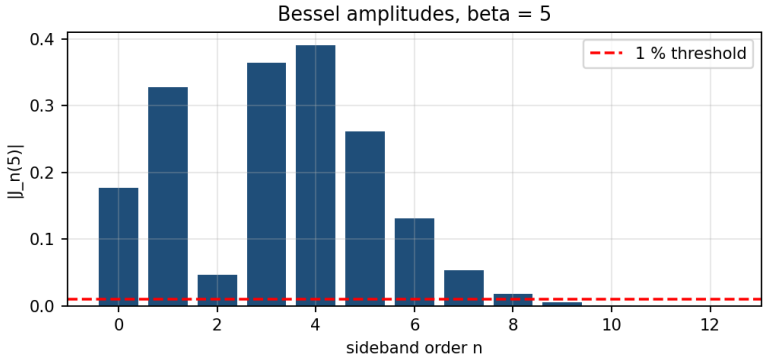


Figure 3: $|J_n(5)|$ – the 1 % criterion (dashed) leaves $n \leq 8$.

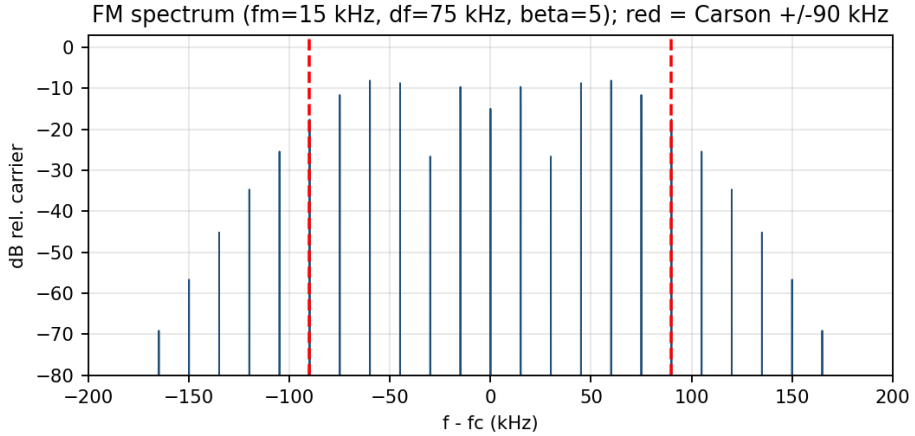


Figure 4: Simulated FM magnitude spectrum for β = 5 (fm = 15 kHz) – the line heights equal $|J_n(5)|$ (see Table); dashed lines are Carson's ±90 kHz.

3.2 Significant sidebands and bandwidth

Method	Result	Bandwidth
Carson's rule	$B_T = 2(\Delta f + W) = 2(75 + 15)$	180 kHz ($n \leq 6$; contains 99.4 % of power)
Significant sidebands, $\lvert J_n \rvert > 0.01$ (1 %)	$n \leq 8$ ($J_8 = 0.0184$, $J_9 = 0.0055$)	$2 \cdot 8 \cdot 15 = 240$ kHz
$\lvert J_n \rvert > 0.05$ (5 %)	$n \leq 7$	$2 \cdot 7 \cdot 15 = 210$ kHz
Universal rule $2f_m(\beta+1)$	same as Carson	180 kHz

Carson's rule captures 99.4 % of the power (the first six pairs). The 1 %-amplitude criterion is more conservative (−40 dB below the unmodulated carrier) and is what an emission mask has to respect. The outer Bessel pairs (n = 7, 8) at ±105 and ±120 kHz are at −25.5 and −34.7 dB, which is inside an FCC-type emission mask (as I recall it: ≥ 25 dB attenuation between 120 and 240 kHz from the carrier and ≥ 35 dB beyond 240 kHz – please verify against the regulation); n = 9 (135 kHz) is at −45 dB.

3.3 Effect of deviation and modulation index

Δf	β = Δf/15 kHz	Carson BW	Adjacent-channel consequence (200 kHz raster)
25 kHz	1.67	80 kHz	Very safe; poor SNR gain ($1.5\beta^2 = 4.2$)
50 kHz	3.33	130 kHz	Safe; SNR gain 16.7
75 kHz	5.0	180 kHz	Fills channel; SNR gain 37.5 (15.7 dB)
100 kHz	6.67	230 kHz	Exceeds 200 kHz; ±115 kHz sidebands spill into adjacent channels
150 kHz	10	330 kHz	Severe adjacent-channel interference

Increasing Δf raises the SNR as β^2 (≈ +6 dB per doubling) but also widens the band approximately in proportion to Δf. Beyond 75 kHz the sideband energy beyond ±100 kHz falls into the neighbouring 200 kHz channels (centres ±200 kHz). **Over-deviation** (loud peaks, pre-emphasis boosting the highs, poorly-set limiters) therefore causes adjacent-channel interference and splatter, and receivers tuned to the neighbouring station hear it as "breakthrough".

3.4 Constant average power

The power is $P = \Sigma J_n^2(\beta) \cdot A_c^2/2 = A_c^2/2$ for every β: increasing deviation moves power from the carrier ($J_0^2 = 3.2$ % at $\beta = 5$, and exactly zero at $\beta = 2.405, 5.52, \ldots$) into the sidebands, without raising total transmitted power. $A_c^2/2$ stays constant, so the PA can run saturated.

3.5 Stated broadcast reference framework

Reference	What it contributes to this design
ITU-R BS.450	Transmission standards for FM sound broadcasting at VHF: maximum deviation ±75 kHz, pre-emphasis 50 or 75 µs, stereo pilot/subcarrier arrangement. Justifies $\Delta f = 75$ kHz and $\tau = 75$ µs.
ITU-R BS.412-9	Planning standards for VHF FM sound broadcasting: channel spacing, protection ratios (wanted-to-unwanted) and field strength. Defines how much co-/adjacent-channel interference the 15 dB C/N margin must tolerate.
IEEE 185-1974	IEEE/IHF Standard Methods of Testing FM Broadcast Receivers (now withdrawn; I verified the title on the IEEE site). Basis for the receiver measurements: sensitivity, selectivity, capture ratio, SNR, distortion, image rejection.

Reference	What it contributes to this design
47 CFR §73.322 (as cited in the brief)	US FCC stereophonic transmission standards (19 kHz pilot, 38 kHz subcarrier, 75 kHz total modulation). The general FM emission mask is, as I recall, in §73.317 – please check both against the current text. Together they fix the 75 kHz limit and the out-of-band emission limits used above.

I summarised ITU-R BS.450, BS.412-9 and 47 CFR §73.322 from general knowledge; only IEEE 185-1974 was checked online. Quote clause numbers from the standards themselves if your instructor requires them.

Case Study Decision 3

I recommend **a peak frequency deviation of 75 kHz (β = 5)** and a **transmission bandwidth of 180 kHz (Carson) within the 200 kHz channel** (with the 1 %-amplitude sidebands extending to ±120 kHz, 240 kHz total, controlled by the emission mask) because it captures 99.4 % of the power, gives +15.7 dB tone-SNR improvement, follows ITU-R/FCC practice, and keeps adjacent-channel spill below the allowed mask. A deviation of 100 kHz was rejected because Carson's bandwidth (230 kHz) exceeds the channel; 50 kHz was rejected because it wastes 6 dB of available SNR gain.

Stage 4 – Transmitter and Receiver Architecture (20 marks)

4.1 Transmitter (audio source to antenna)

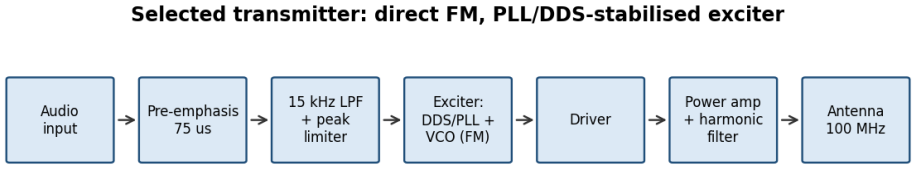


Figure 5: Selected transmitter – direct FM with PLL/DDS-stabilised exciter.

Chain: studio audio → 75 µs pre-emphasis → 15 kHz low-pass and peak deviation limiter (prevents $|\Delta f| > 75$ kHz) → exciter (VCO/DDS modulated at 100 MHz, locked by a PLL to a crystal reference) → driver → Class-C/E power amplifier → harmonic/low-pass filter and emission-mask filter → antenna.

4.2 FM and PM generation methods

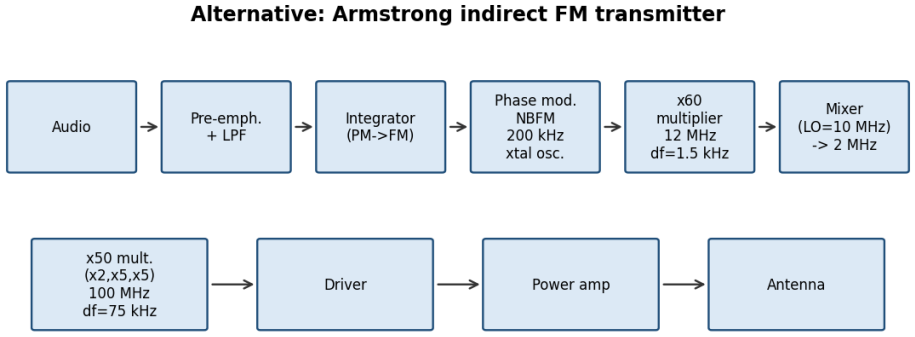


Figure 6: Armstrong indirect FM transmitter (alternative).

Method	Principle	Numbers (this design)	Pros / cons
Direct FM	Message varies a VCO / varactor frequency; PLL holds centre frequency	VCO at 100 MHz, $k_f = 75$ kHz/V; PLL loop bandwidth ≪ 30 Hz so that audio is not suppressed	+ Large deviation directly, simple; − needs frequency stabilisation (PLL/AFC); loop must not remove audio
Indirect (Armstrong)	Integrate m(t) → narrow-band PM with crystal oscillator; multiply and mix to get wideband FM	Crystal 200 kHz, $\Delta f_i = 25$ Hz; ×60 → 12 MHz ($\Delta f = 1.5$ kHz); mix with 10 MHz → 2 MHz; ×50 → 100 MHz, $\Delta f = 75$ kHz ($n_{total} = 3000$)	+ Very stable (crystal); − many stages, multiplies noise/phase error, distortion in low-β phase modulator
PM generation	Phase modulator driven by m(t) (or FM by integrating m first)	—	Used inside Armstrong; PM is simple and crystal-stable

Selected: modern direct FM with a PLL/DDS exciter. It gives 75 kHz deviation in one step, excellent linearity, and a crystal-referenced carrier (±1 ppm = ±100 Hz), while avoiding the cumulated distortion and large multiplier chain of Armstrong's method. Armstrong is kept as a documented alternative.

4.3 Receiver architecture

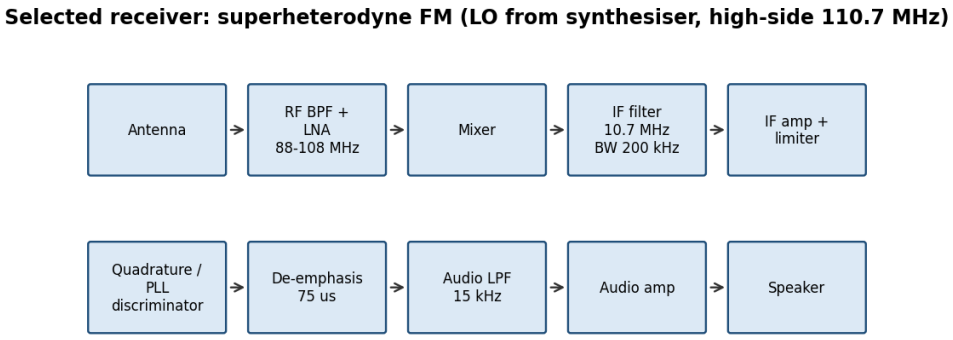


Figure 7: Selected receiver – superheterodyne FM with limiter and discriminator.

4.4 Receiver architecture and IF selection

Option	Assessment
TRF (tuned RF)	Poor selectivity at 100 MHz, unstable gain – rejected.
Direct conversion	Needs DC-coupled I/Q path; difficult for 200 kHz FM at VHF without DSP – rejected for the basic receiver.
Superheterodyne (selected)	Fixed IF gives constant selectivity; mixer + LO tunes the station.
IF = 455 kHz	Image only 0.9 MHz away – inadequate for 200 kHz channels at 100 MHz; and the IF is too low for 150–200 kHz FM bandwidth.
IF = 10.7 MHz (selected)	Industry standard; image at $f_c + 2 \cdot IF$ = 121.4 MHz, 21.4 MHz away, easily removed by the RF band-pass filter; ceramic/crystal 200 kHz IF filters are cheap.

With high-side injection $f_{LO} = f_c + IF$ = 110.7 MHz and image $f_{im} = f_c + 2 \cdot IF$ = 121.4 MHz (outside the 88–108 MHz band, so the RF filter can reject it by > 60 dB).

4.5 Limiter and discriminator

Limiter: a high-gain, hard-limiting IF amplifier that removes envelope variations caused by noise, fading and AM interference, supplying a constant-amplitude signal to the detector. It also provides "capture effect" (the stronger of two co-channel stations dominates). Its dynamic range (≈ 60–80 dB) sets the C/N at which limiting is complete. **Discriminator:** a circuit whose output voltage is proportional to the deviation of the instantaneous frequency from 10.7 MHz – e.g. a quadrature detector (limited IF multiplied by its phase-shifted copy and low-passed), a PLL demodulator (VCO control voltage follows m(t)) or a differentiator + envelope detector. The differentiator makes the output $\propto f_i(t) = f_c + k_f m(t)$; noise at its output has a parabolic power spectrum.

4.6 Receiver parameters

Parameter	Requirement / calculation	Design value
Sensitivity	Noise floor in 200 kHz: −121 dBm; NF = 5 dB → −116 dBm; required C/N = 15 dB	≈ −101 dBm for 15 dB IF C/N (≈ −106 dBm at the 10 dB threshold)
Selectivity	Pass ±100 kHz; reject adjacent channel at ±200 kHz and alternate at ±400 kHz	IF filter 200 kHz (−3 dB); ≥ 40 dB at ±200 kHz, ≥ 60 dB at ±400 kHz
Image rejection	Image at 121.4 MHz	≥ 60 dB from RF BPF/LNA preselector
Fidelity	Reproduce 30 Hz–15 kHz	±1 dB audio response after de-emphasis; THD < 1 %

Parameter	Requirement / calculation	Design value
Stability	LO accuracy: 1 ppm of 110.7 MHz = 111 Hz ≪ 100 kHz	Synthesised LO (PLL), ±2 kHz total; no AFC needed
Adjacent-channel rejection	Capture ratio 1–2 dB; protection ratio per ITU-R BS.412	Limiter + narrow IF filter
AM rejection	Limiter removes envelope noise	≥ 40 dB

Case Study Decision 4

I recommend a **direct-FM, PLL/DDS-stabilised transmitter with 75 µs pre-emphasis, and a superheterodyne receiver with 10.7 MHz IF, a 200 kHz IF filter, a hard limiter and a quadrature/PLL discriminator** because they give a stable crystal-referenced carrier, deviation of 75 kHz in a single stage, sensitivity of ≈ −101 dBm, ≥ 60 dB image rejection, ≥ 40 dB adjacent-channel rejection and a flat 15 kHz audio response, at low complexity and cost. Armstrong indirect FM would be acceptable but needs 3000× multiplication and more stages; a TRF or 455 kHz-IF receiver would not meet selectivity/image requirements.

Stage 5 – AM and DSB-SC Noise Performance (15 marks)

5.1 Noise model and assumptions

AWGN with two-sided PSD $N_0/2$, band-pass filtered to the signal bandwidth. Narrow-band representation: $n(t) = n_I(t)\cos 2\pi f_c t - n_Q(t)\sin 2\pi f_c t$; $E[n_I^2] = E[n_Q^2] = E[n^2] = 2N_0W$ for a double-sideband signal of message bandwidth W (double-sided bandwidth 2W). Receiver ideal, perfect synchronisation unless noted, message power $P = E[m^2]$. **Reference channel SNR:** $\gamma = P_T/(N_0W)$, where P_T is the transmitted power (the SNR of baseband transmission with the same power in bandwidth W).

5.2 AM (envelope detection)

$$x(t) = A_c[1 + k_a m(t)]\cos 2\pi f_c t + n(t)$$

For large carrier-to-noise ratio the envelope is $y(t) \approx A_c[1 + k_a m(t)] + n_I(t)$. The DC term is blocked, giving signal power $A_c^2 k_a^2 P$ and noise power $2N_0W$:

$$SNR_{o,AM} = A_c^2 k_a^2 P / (2N_0W); \quad P_T = A_c^2(1 + k_a^2 P)/2; \quad SNR_{o,AM} = [k_a^2 P/(1 + k_a^2 P)]\cdot\gamma$$

For tone modulation with index μ: figure of merit $= \mu^2/(2 + \mu^2)$, **1/3 (−4.77 dB) at μ = 1**; at μ = 0.3 (typical for speech): 0.043 (−13.7 dB).

5.3 DSB-SC (coherent detection)

$$x(t) = A_c m(t)\cos 2\pi f_c t + n(t); \text{ after multiplier \& LPF: } y(t) = (1/2)[A_c m(t) + n_I(t)]$$

$$SNR_{o,DSB} = A_c^2 P/(2N_0W); \quad P_T = A_c^2 P/2; \quad SNR_{o,DSB} = \gamma \text{ (figure of merit = 1)}$$

If the local oscillator has a phase error φ the output signal is multiplied by cos φ, so SNR falls by $\cos^2\varphi$: 5° → −0.03 dB; 10° → −0.13 dB; 30° → −1.25 dB; 90° → no output (quadrature null). A frequency error produces beating at the difference frequency. Hence DSB-SC needs carrier recovery (squaring loop/Costas loop or a transmitted pilot).

5.4 Calculation sheet

Quantity	AM (μ = 1)	AM (μ = 0.3)	DSB-SC	FM β = 5 (tone, above threshold)
Figure of merit (SNR_o/γ)	0.333	0.043	1	37.5
In dB	−4.77	−13.7	0	+15.7
Output SNR at γ = 10 dB	5.2 dB	−3.7 dB	10 dB	25.7 dB
Output SNR at γ = 30 dB	25.2 dB	16.3 dB	30 dB	45.7 dB
Output SNR at γ = 40 dB	35.2 dB	26.3 dB	40 dB	55.7 dB
Carrier power as % of P_T (tone)	66.7 % (μ=1)	95.7 %	0 %	—
Detector	Envelope (simple)	Envelope	Coherent (sync needed)	Limiter + discriminator
Complexity	Lowest	Lowest	Moderate–high	Moderate
Bandwidth	2W = 30 kHz	30 kHz	30 kHz	180 kHz

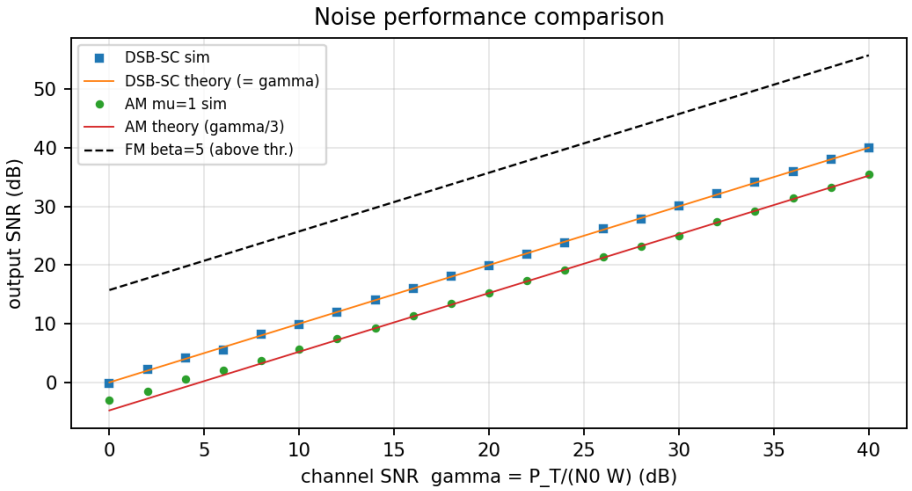


Figure 8: Simulated output SNR versus channel SNR γ. Markers = simulation; lines = theory. FM line is valid only above threshold.

Simulation check: the DSB-SC simulation matches SNR_o = γ within ≈ 0.3 dB over 0–40 dB, and the AM simulation matches γ/3 at high γ (35.5 dB at γ = 40 dB versus 35.2 dB). At low γ the AM envelope detector departs from the linear theory (threshold effect of the envelope detector), as expected.

5.5 Engineering interpretation

- **Power efficiency:** DSB-SC spends all power on the sidebands and reaches the best SNR of any linear system (SNR_o = γ). Standard AM wastes ≥ 67 % of power on the carrier – its advantage is only a cheap envelope detector.
- Neither linear scheme can trade bandwidth for SNR: output SNR ≤ γ, so a 30 dB output would need 30 dB channel SNR.
- FM, with the same transmitted power, gives +15.7 dB over DSB-SC (tone) at the cost of 6× bandwidth (180 kHz vs 30 kHz) – a spectrum the broadcast allocation already provides.

Stage 5 Decision

I recommend using **DSB-SC (SNR_o = γ, the best linear-modulation noise performance) as the reference/noise-performance benchmark** because it uses all of its power for information and reaches the upper bound for linear modulation; AM is shown to be 4.77 dB worse at μ = 1. The comparison shows that FM beats the best linear system by 15.7 dB (tone) at the same transmitted power, confirming that the extra bandwidth of FM is worth spending and justifying the selection of FM in Stage 2; the price is a threshold (Stage 6) that linear systems do not have.

Stage 6 – FM Noise, Pre-/De-emphasis and Threshold (25 marks)

6.1 FM noise mechanism and limiter/discriminator operation

At the receiver, $x(t) = A_c \cos[2\pi f_c t + \varphi(t)] + n(t)$. For large carrier-to-noise ratio, the phase of the sum is $\varphi(t) + n_Q(t)/A_c$, i.e. noise perturbs only the phase component in quadrature with the carrier; the limiter removes the amplitude component of the noise entirely. The discriminator differentiates the phase, so the noise at its output is the derivative of the quadrature noise: **noise PSD = $(N_0/A_c^2)\, f^2$**, i.e., parabolic – small at low audio frequencies and large near 15 kHz. The signal output is $k_f \cdot m(t)$ (proportional to the deviation).

6.2 Output SNR relation

$$SNR_{o,FM} = 3 A_c^2 k_f^2 P/(2N_0W^3) = 3\beta_W^2 (P/m_{max}^2)\, \gamma, \quad \beta_W = \Delta f/W$$

For a sinusoidal message ($P/m_{max}^2 = ½$) this is **$SNR_o = 1.5\beta^2\gamma$**, where $\gamma = A_c^2/(2N_0W)$. At $\beta = 5$: $37.5\gamma \rightarrow$ **+15.7 dB over baseband** (relative to baseband transmission with the same power; DSB-SC equals baseband, $SNR_o = \gamma$). For real program material the average-to-peak power ratio is lower than ½ (typically 0.1–0.2), so the improvement is smaller than the tone value ($3\beta^2 \cdot 0.1 = 7.5$, i.e. 8.8 dB). The IF C/N and γ are related by $\gamma = (C/N) \cdot (B_{IF}/W)$: for $B_{IF} = 200$ kHz this adds +11.25 dB.

6.3 Effect of frequency deviation on SNR and bandwidth

Δf (kHz)	IF filter used (kHz)	IF C/N at fixed γ = 35 dB (dB)	Simulated SNR_o (dB)	Theory 1.5β²γ (dB)
10	40.0	30.7	33.8	33.2
25	57.2	29.2	39.4	41.2
50	112.2	26.3	46.6	47.2
75	167.2	24.5	49.8	50.7
100	222.2	23.3	51.3	53.2

SNR rises ≈ 6 dB per doubling of Δf (theory and simulation agree within 0.5–1.9 dB; the larger gap at $\Delta f = 100$ kHz is due to the finite IF filter). At fixed N_0 the IF C/N drops 7 dB between $\Delta f = 10$ and 100 kHz because the noise bandwidth is wider: deviation therefore cannot be increased indefinitely – the system moves closer to threshold and out of the channel allocation.

6.4 Threshold effect

The linear relation $SNR_o = 1.5\beta^2\gamma$ holds only when carrier ≫ noise. When C/N falls to about 10 dB the noise phasor occasionally exceeds the carrier and encircles the origin; each such event produces a 2π phase step, i.e. an impulse ("click") at the discriminator output. The clicks add a broadband noise whose rate increases rapidly as C/N drops, so SNR_o falls quickly (≈ 3–5 dB per dB of C/N). Rule of thumb: threshold at $\gamma_{th} \approx 20(\beta + 1) = 120$ (20.8 dB), i.e. $C/N_{IF} = \gamma_{th} \cdot W/B_T = 120/12 = 10$ dB.

IF C/N (dB)	γ (dB)	Theory 1.5β²γ (dB)	Simulated SNR_o (dB)	Condition
0	11.3	27.0	5.0	Below threshold (clicks)
3	14.3	30.0	10.8	Below threshold (clicks)
5	16.3	32.0	15.2	Below threshold (clicks)
6	17.3	33.0	19.5	Below threshold (clicks)
8	19.3	35.0	28.3	Near threshold
10	21.3	37.0	36.7	Near threshold
12	23.3	39.0	39.2	Moderate

IF C/N (dB)	γ (dB)	Theory 1.5β²γ (dB)	Simulated SNR_o (dB)	Condition
15	26.3	42.0	41.8	Moderate
20	31.3	47.0	46.5	High C/N
25	36.3	52.0	50.5	High C/N
30	41.3	57.0	53.6	High C/N

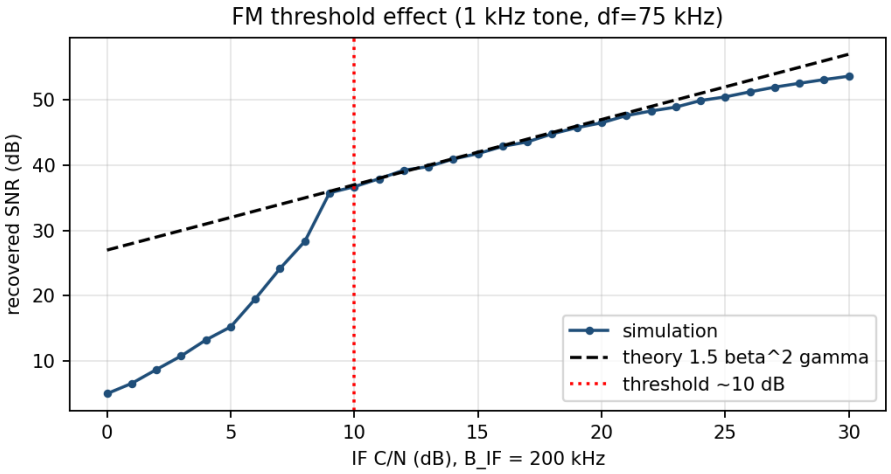


Figure 9: FM threshold effect – simulated recovered SNR versus IF C/N (1 kHz tone, Δf = 75 kHz). The knee is at ≈ 10 dB.

Reading the plot: from 30 dB down to 12 dB the simulated curve follows the 1:1 dB-per-dB straight line within ≈ 0.5 dB; at 10 dB it just touches the theory (36.7 dB vs 37.0 dB); at 8 dB it is already 6.7 dB below, and at 5 dB it is 17 dB below the linear prediction (15.2 dB instead of 32 dB). This is the threshold – FM trades abrupt failure for the large SNR improvement above it.

6.5 Pre-emphasis and de-emphasis

Because the discriminator noise is ∝ f² while typical audio energy falls with frequency, the high-frequency end of the audio band has the worst SNR. The transmitter boosts the highs before modulation and the receiver cuts them by the exact inverse after demodulation:

$$H_{pe}(f) = 1 + j\,f/f_1, \quad H_{de}(f) = 1/(1 + j\,f/f_1), \quad f_1 = 1/(2\pi\tau) = 2.12\ \text{kHz}\ (\tau = 75\ \mu\text{s})$$

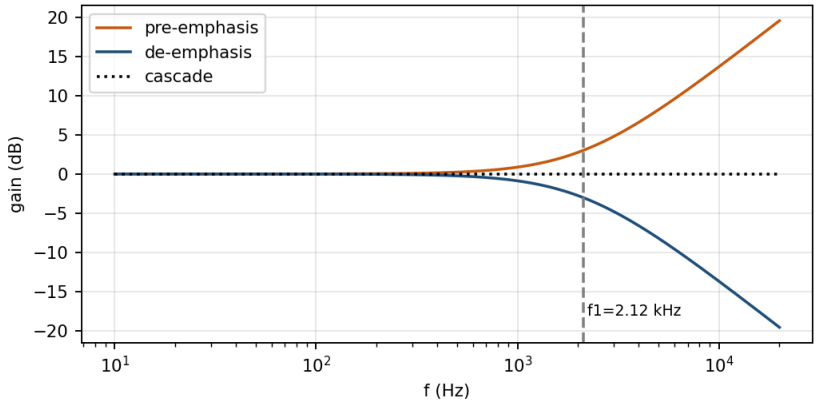


Figure 10: Pre-emphasis and de-emphasis responses (cascade = 0 dB: the audio is not altered).

The cascade is flat, so the audio is unchanged, but the de-emphasis filter attenuates the discriminator noise (which was added after pre-emphasis) at high frequencies. Ideal improvement for a message whose spectrum is flattened by pre-emphasis:

$$I = (W/f_1)^3 / \{3[(W/f_1) - \arctan(W/f_1)]\}$$

With $W/f_1 = 7.07$: $I = 20.9 =$ **13.2 dB** (ideal). With 50 µs ($f_1 = 3.18$ kHz, $W/f_1 = 4.71$) the ideal figure is ≈ 10.2 dB.

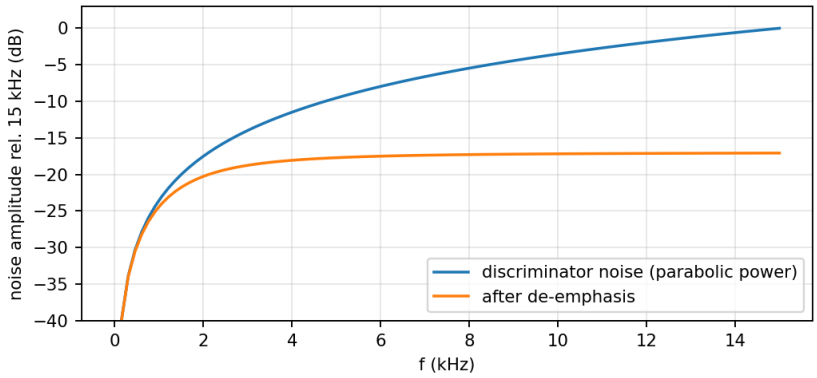


Figure 11: Discriminator noise spectrum before and after de-emphasis (amplitude relative to the value at 15 kHz).

6.6 Simulation with and without pre-/de-emphasis

Two test messages with identical rms deviation (25 kHz rms ≈ 75 kHz at 3σ) were used:

- **Flat-spectrum 0–15 kHz message.** Pre-emphasis would raise the HF deviation, so the whole signal has to be scaled down to keep the same rms deviation – the net effect is theoretically $10\log[I/(1 + (W/f_1)^2/3)] = +0.7$ dB. Simulation: −0.2 … +0.1 dB (within the noise of one realisation). Pre-emphasis gives essentially **no** benefit for a flat spectrum – an important caution.
- **Audio-like message** (spectrum falling like the de-emphasis response, as for speech/music). Pre-emphasis flattens the transmitted spectrum and the scaling penalty is small ($\sigma_{pe} = 2.33$): theoretical gain 5.9 dB.

IF C/N (dB)	SNR without (dB)	SNR with 75 µs (dB)	Gain (dB)
30	42.4	43.0	0.5
25	40.9	42.4	1.5
20	38.7	41.6	2.9
15	34.4	39.5	5.0
12	32.2	36.9	4.6
10	30.0	35.4	5.4

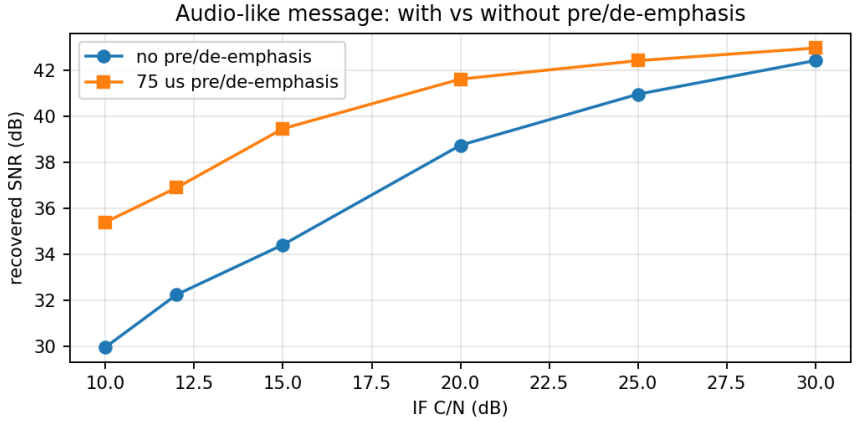


Figure 12: Audio-like message – recovered SNR with and without pre-/de-emphasis.

At C/N ≤ 20 dB the simulated improvement is 2.9–5.4 dB, close to the theoretical 5.9 dB for this message; at 30 dB the gain shrinks to 0.5 dB because both curves are limited by the distortion of the 200 kHz IF filter and the discretisation of the discriminator (≈ 56 dB noiseless floor), not by noise. The ideal 13.2 dB applies to a message whose own spectrum is as strong at high frequencies as at low ones **and** has been scaled to the same peak deviation, which real programme material rarely satisfies; in practice broadcasters measure ≈ 8–12 dB with limiting/compression of the highs.

6.7 Trade-off: noise improvement, audio response, bandwidth and complexity

Factor	Effect
Noise improvement	+5.9 dB (audio-like message, simulated/theory) up to +13.2 dB (ideal); costs 1 RC network at each end
Audio response	Flat after de-emphasis only if both τ match (75 µs both sides); mismatch gives treble boost/cut
Bandwidth	HF boost raises deviation at 5–15 kHz; a deviation limiter/compressor is required to keep $\|\Delta f\| \leq 75$ kHz and the Carson bandwidth ≤ 180 kHz; over-deviation → distortion and adjacent-channel splatter
Complexity	Very low (passive RC or one op-amp); essential in all broadcasts

6.8 Operating C/N and strategy

Threshold is at C/N ≈ 10 dB. To allow for fading, interference and receiver tolerance a margin of ≥ 5 dB is required. At the recommended minimum IF C/N = 15 dB: tone SNR = 15 + 11.25 + 15.74 = **42 dB** (simulation: 41.8 dB); adding the 5.9 dB pre-/de-emphasis gain for audio-like programme gives ≈ **48 dB**. At 20 dB: 47 dB (sim 46.5) → ≈ 53 dB with emphasis. Operation above 25–30 dB gives diminishing returns because the SNR is then limited by the system distortion (≈ 54–56 dB) rather than by noise.

Stage 6 Decision

I recommend a **75 µs pre-emphasis / 75 µs de-emphasis pair (f_1 = 2.12 kHz) with a peak-deviation limiter set at 75 kHz** and an **operating IF C/N of approximately 15–20 dB (minimum design value 15 dB, 5 dB above the ≈ 10 dB threshold)** because at 15 dB the simulated tone SNR is 41.8 dB (theory 42.0 dB) and ≈ 48 dB with emphasis; at 10 dB it is 36.7 dB and at 5 dB it collapses to 15–17 dB (threshold). 50 µs would be selected only to follow European practice; no pre-emphasis would forfeit about 6–13 dB.

Stage 7 – MATLAB-Based Verification and Critical Evaluation (15 marks)

Method. All signals are processed as complex-envelope (baseband-equivalent) sequences at fs = 2 MHz, N = 65 536 samples: FM modulation by FFT integration of m(t); AWGN added with the complex variance set so that the C/N in a 200 kHz brick-wall IF filter is the requested value; limiter/discriminator implemented as angle(x[n]x*[n−1])·fs/2π; 15 kHz audio low-pass; SNR measured against the clean message. The MATLAB script (Appendix A) implements the same steps; the figures below came from the equivalent Python run.

7.1 Message, FM and PM waveforms

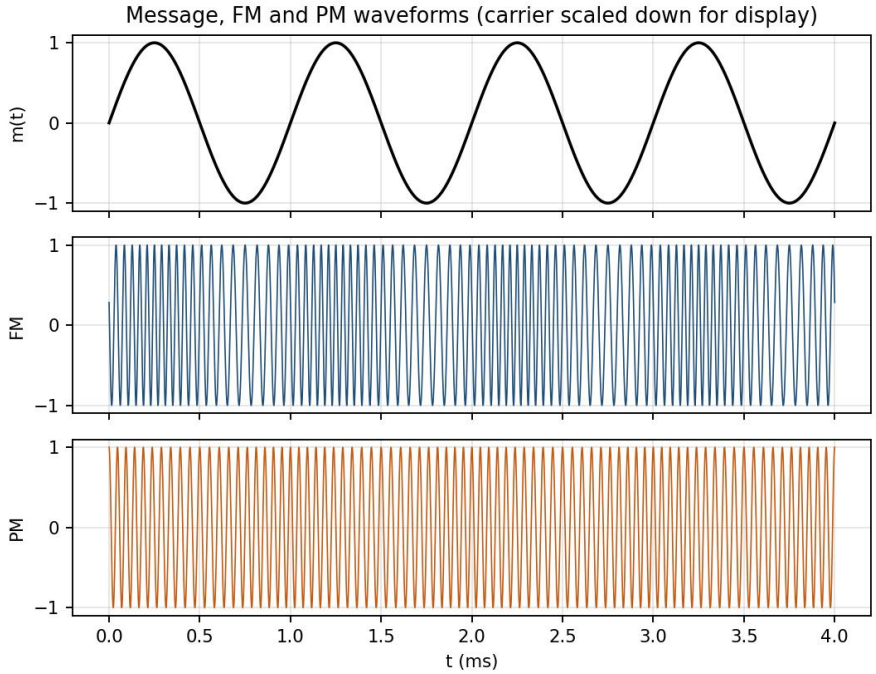


Figure 13: Message, FM (β = 5) and PM (β_p = 1.5) waveforms (carrier scaled to 20 kHz for display).

Interpretation: the FM waveform has constant amplitude and the local frequency is highest at the positive message peaks (compressed cycles) and lowest at the negative peaks. In the PM waveform the instantaneous frequency is highest where the message slope is steepest (the zero crossings). Both have unit envelope – constant power.

7.2 Spectrum and bandwidth verification

n	$\|J_n(5)\|$ theory	Simulated line level	Difference
0	0.17760	0.17760	2.3e-12
1	0.32758	0.32758	3.0e-12
2	0.04657	0.04657	4.7e-12
3	0.36483	0.36483	1.3e-11
4	0.39123	0.39123	1.9e-11
5	0.26114	0.26114	1.8e-11
6	0.13105	0.13105	1.3e-11
7	0.05338	0.05338	8.0e-12
8	0.01841	0.01841	4.5e-12
9	0.00552	0.00552	2.5e-12

Simulated line levels from the FFT of the FM signal agree with the Bessel amplitudes to better than 10^{-9}, **verifying $\beta = \Delta f / f_m = 5$**. Line levels at n = 7, 8, 9 (0.053, 0.018, 0.0055) confirm that significant sidebands extend to n = 8 (1 % rule, 240 kHz), whilst the cumulative power within the Carson band ($n \leq 6$) is 99.36 % – so the theoretical 180 kHz Carson bandwidth is verified (Figure 4).

7.3 Noisy received signals and recovered audio

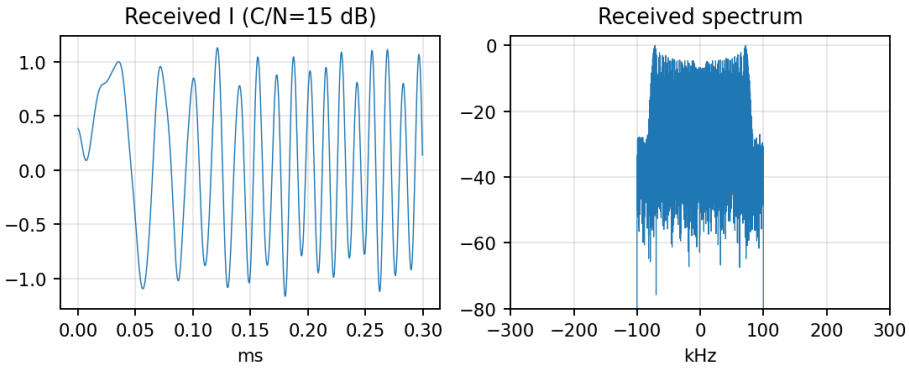


Figure 14: Received FM signal (I-component) and spectrum at C/N = 15 dB.

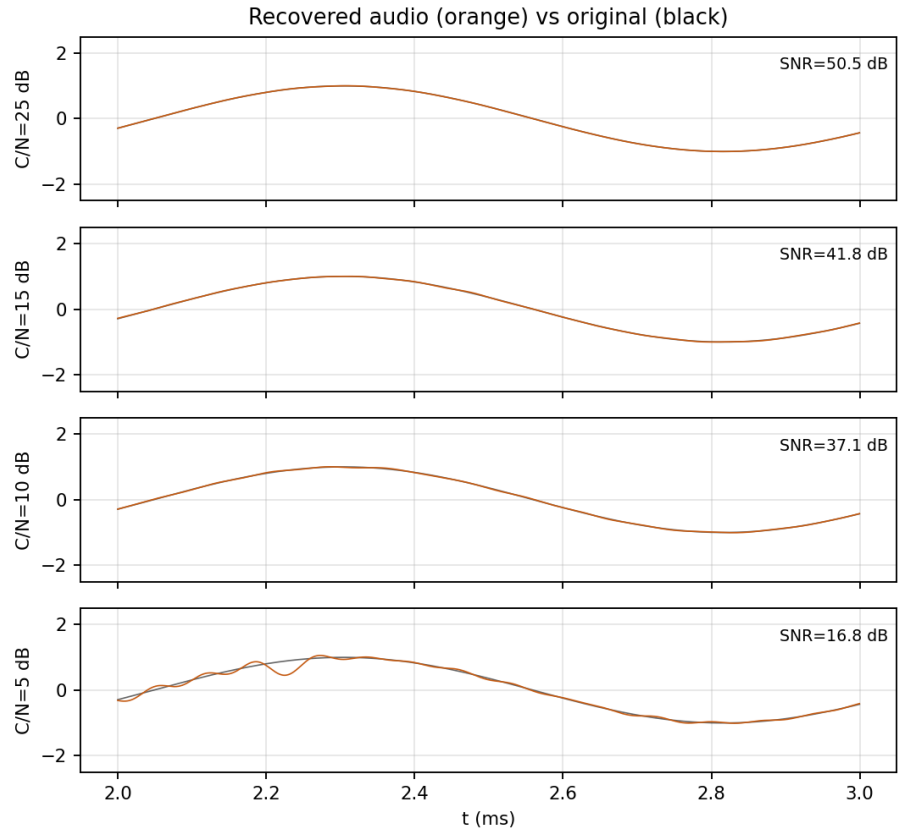


Figure 15: Recovered audio at high (25 dB), moderate (15 dB), near-threshold (10 dB) and below-threshold (5 dB) C/N.

Condition	C/N (dB)	Recovered SNR (dB)	Interpretation
High	25	50.5	Clean sine; noise barely visible
Moderate	15	41.8	Small hiss; meets design goal (≥ 40 dB)
Near threshold	10	37.1	Noticeable noise; clicks start to appear
Below threshold	5	16.8	Frequent spikes (clicks) – crackling, unacceptable

7.4 Threshold effect

See Figure 9 and Section 6.4: the simulated curve follows $1.5\beta^2\gamma$ above C/N ≈ 12 dB, deviates sharply below 10 dB. The measured knee at ≈ 10 dB agrees with the rule $\gamma_{th} = 20(\beta + 1)$.

7.5 Pre-/de-emphasis comparison

See Section 6.6, Figure 12. Gain 2.9–5.4 dB at C/N = 10–20 dB for audio-like programme; ideal 13.2 dB. For the flat-spectrum test there is no gain, as predicted.

7.6 AM and DSB-SC repeat

See Figure 8 and Section 5.4: simulation reproduces $SNR_o = \gamma$ for DSB-SC and $\gamma/3$ for AM ($\mu = 1$) at high γ. FM above threshold is 15.7 dB better than DSB-SC and 20.5 dB better than AM.

7.7 Theory versus simulation – discrepancies and engineering significance

Observation	Likely cause	Engineering significance
At C/N ≥ 25 dB the simulated SNR saturates (53.6 dB at 30 dB vs 56.9 dB theory; noiseless floor 56.3 dB)	Truncation of FM sidebands by the 200 kHz brick-wall IF filter and discrete-time differentiation	Real IF filters and the demodulator limit audio SNR to ≈ 55–60 dB; raising C/N above ≈ 25 dB buys nothing – no need for more transmitter power
Below ≈ 10 dB, simulation lies far under the $1.5\beta^2\gamma$ line (−6.7 dB at 8 dB, −17 dB at 5 dB)	Threshold: click noise, not modelled by the linear theory	Defines the coverage limit; the design margin of 5 dB above threshold is essential
Pre-emphasis gain 5.4 dB vs 13.2 dB ideal	Ideal formula assumes a message whose spectrum is flattened by pre-emphasis without extra deviation; real/test messages and the deviation limit reduce the gain	Do not promise the textbook 13 dB; budget 5–10 dB for programme material
Flat-spectrum test: ≈ 0 dB gain	Pre-emphasis forces lower overall gain to hold deviation	Pre-emphasis must be matched to programme spectrum; audio processing (limiter/compressor) is part of the design
Sim vs theory 0.5–1.9 dB at Δf = 25–100 kHz (Table, 6.3)	Finite IF bandwidth, single noise realisations	Within normal measurement tolerance

Stage 7 Decision

Based on the verification, I recommend the **final configuration: FM, f_c = 100 MHz, Δf = 75 kHz (β = 5), Carson bandwidth 180 kHz in a 200 kHz channel, 75 µs pre-/de-emphasis with deviation limiter, superheterodyne receiver (10.7 MHz IF, 200 kHz filter, limiter + quadrature/PLL discriminator) and a design IF C/N of 15–20 dB** because the simulation verified β, the Bessel spectrum and bandwidth exactly, reproduced the SNR formula within ≈ 0.5 dB above threshold and located the threshold at ≈ 10 dB. The significant discrepancies (distortion floor, threshold, reduced pre-emphasis gain) all indicate real engineering limits and are included in the margins.

Final Engineering Decision

WOULD YOU DEPLOY THIS BROADCAST COMMUNICATION SYSTEM? → YES
Conditional YES – deploy the design for mono audio, with the conditions listed below.

Defence of the decision

- **Modulation selection:** FM beats PM by 4.77 dB at equal peak deviation, uses the 200 kHz channel and keeps the receiver simple (Stage 2).
- **Spectrum/bandwidth compliance:** Carson 180 kHz holds 99.4 % of the power; the 1 % components stay within ±120 kHz, so adjacent-channel interference is controlled provided the deviation limiter holds 75 kHz (Stage 3).
- **Constant average power:** envelope is constant, power $A_c^2/2$ regardless of programme, so a saturated efficient PA can be used.
- **Architecture:** direct, PLL-stabilised exciter and superheterodyne receiver with 10.7 MHz IF give ≈ −101 dBm sensitivity at C/N 15 dB, ≥ 60 dB image and ≥ 40 dB adjacent-channel rejection (Stage 4).
- **AM/DSB-SC comparison:** best linear system gives $SNR_o = \gamma$; FM gives +15.7 dB at the same power (Stage 5).
- **FM noise and threshold:** verified recovered SNR 41.8 dB at 15 dB C/N, 36.7 dB at 10 dB, threshold at ≈ 10 dB (Stage 6, 7).
- **Pre-/de-emphasis:** 75 µs pair adds 5.9 dB (simulated for audio-like material; 13.2 dB ideal) at negligible cost.

Conditions and what to monitor (so that the YES stays valid)

- The coverage contour must be set where IF C/N ≥ 15 dB (field strength for ≈ −101 dBm at the receiver). Below ≈ 10 dB the service fails abruptly (clicks).
- Peak deviation must be hard-limited to 75 kHz after pre-emphasis; check with a modulation monitor.
- This analysis is mono only. A 19 kHz pilot and 38 kHz stereo subcarrier (or RDS) widen the baseband to 53–57 kHz and reduce the stereo SNR by ≈ 20 dB; in that case C/N must be raised or the stereo coverage contour reduced.
- If multipath or heavy interference are expected, increase the design margin (C/N 20 dB).

Final specification-verification table

Requirement	Specified	Achieved	Satisfied?
Message bandwidth	15 kHz	15 kHz (audio LPF)	YES
Carrier frequency	100 MHz	100 MHz (PLL/DDS, ±1 ppm)	YES
Peak deviation	75 kHz / selected	75 kHz (limiter)	YES
FM modulation index	Calculated	$\beta = \Delta f/W = 5.0$ (verified by Bessel lines)	YES
FM bandwidth	Calculated	Carson 180 kHz (< 200 kHz channel); 1 % sidebands 240 kHz	YES
Noise condition	AWGN/interference	AWGN C/N 0–30 dB simulated; ≥ 15 dB operating	YES
Pre/de-emphasis	Selected	75 µs pair, f_1 = 2.12 kHz; +5.9 dB simulated, 13.2 dB ideal	YES

Requirement	Specified	Achieved	Satisfied?
Receiver architecture	Selected	Superhet, 10.7 MHz IF, 200 kHz filter, limiter + discriminator	YES
MATLAB verification	Required	MATLAB script supplied (Appendix A); numbers/figures from equivalent Python run	PARTIAL – re-run in MATLAB before submitting

If a redesign were needed (for reference)

If field tests show C/N < 12 dB at the contour, the options in order of preference are: (1) raise transmitter power or antenna gain (no bandwidth cost); (2) improve the receiver front end (lower noise figure, better antenna); (3) reduce the deviation to 50 kHz ($\beta = 3.33$, Carson 130 kHz): this narrows the channel and reduces adjacent-channel interference but costs 3.5 dB of SNR ($20\log(50/75)$) and, with the narrower IF filter, lowers the noise bandwidth, so it is a last resort. A change of carrier frequency is not needed because the 88–108 MHz allocation is fixed.

Appendix A – MATLAB Verification Script

Not executed in the environment used to write this report – run it in MATLAB (R2018b or later) and check the figures against those in the report.

```
%% ECA08 CO2 - FM broadcast case study: MATLAB verification package
%  Requires R2018b+ (yline). Signal Processing Toolbox NOT required.
clear; clc; close all; rng(1);
W=15e3; Df=75e3; fc=100e6; beta=Df/W; tau=75e-6; f1=1/(2*pi*tau);

%% 1. Bessel sidebands and bandwidth (Stage 3)
n=0:12; Jn=besselj(n,beta);
nsig=max(n(abs(Jn)>0.01));
BW_carson=2*(Df+W); BW_sig=2*nsig*W;
pw=arrayfun(@(k) sum(besselj(-k:k,beta).^2),0:9);
fprintf('beta=%.2f | Carson=%.0f kHz | 1%% sidebands n=%d -> %.0f kHz | power in 6 pairs=%.4f\n',...
        beta,BW_carson/1e3,nsig,BW_sig/1e3,pw(7));
figure; stem(n,abs(Jn),'filled'); yline(0.01,'r--'); grid on
xlabel('sideband order n'); ylabel('|J_n(5)|'); title('Bessel amplitudes, \beta=5');

%% 2. Simulation grid (complex baseband, carrier removed)
fs=2e6; N=2^16; t=(0:N-1)'/fs;
f=[0:N/2-1, -N/2:-1]'*fs/N;            % FFT frequency axis
Bif=200e3;                             % IF bandwidth
fm=32*fs/N;                            % ~1 kHz tone, integer cycles in record
m=sin(2*pi*fm*t);

%% 3. FM and PM waveforms (carrier scaled down for display only)
td=(0:1/fs:4e-3)'; fcd=20e3; fmd=1e3;
sFM=cos(2*pi*fcd*td-5*cos(2*pi*fmd*td));
sPM=cos(2*pi*fcd*td+1.5*sin(2*pi*fmd*td));
figure; subplot(3,1,1); plot(td*1e3,sin(2*pi*fmd*td)); ylabel('m(t)'); title('Message, FM, PM');
subplot(3,1,2); plot(td*1e3,sFM); ylabel('FM'); subplot(3,1,3); plot(td*1e3,sPM); ylabel('PM'); xlabel('t (ms)');

%% 4. FM spectrum, beta=5 (fm=15 kHz tone)
fs2=3.84e6; N2=2^17; t2=(0:N2-1)'/fs2; x=exp(1j*beta*sin(2*pi*W*t2));
w=hann(N2); X=fftshift(fft(x.*w))/sum(w); f2=(-N2/2:N2/2-1)'*fs2/N2;
figure; plot(f2/1e3,20*log10(abs(X)+1e-9)); xlim([-200 200]); ylim([-80 3]); grid on
xline(-90,'r--'); xline(90,'r--'); xlabel('f-f_c (kHz)'); ylabel('dB'); title('FM spectrum (red = Carson)');
% NOTE: hann() is in Signal Processing Toolbox; replace by 0.5-0.5*cos(2*pi*(0:N2-1)'/N2) if absent.

%% 5. FM link: noise, discriminator, threshold (Stage 6/7)
xc=fmmod(m,Df,f,fs);
cn=0:1:30; snrFM=zeros(size(cn));
for k=1:numel(cn)
   r=zeros(1,3);
   for j=1:3
      y=addnoise(xc,cn(k),Bif,f,fs);
      d=lpf(disc(y,fs)/Df,W,f); r(j)=snrdb(m,d);
   end
   snrFM(k)=mean(r);
end
gam=cn+10*log10(Bif/W); theory=gam+10*log10(1.5*beta^2);
figure; plot(cn,snrFM,'o-',cn,theory,'k--'); grid on; xline(10,'r:');
xlabel('IF C/N (dB)'); ylabel('recovered SNR (dB)'); legend('simulation','1.5\beta^2\gamma','threshold');
title('FM threshold effect');

% Recovered audio at high / moderate / near-threshold / below-threshold
figure;
for k=1:4
   c=[25 15 10 5](k); y=addnoise(xc,c,Bif,f,fs); d=lpf(disc(y,fs)/Df,W,f);
   subplot(4,1,k); plot(t(4001:6000)*1e3,m(4001:6000),'k',t(4001:6000)*1e3,d(4001:6000),'r');
   ylabel(sprintf('C/N=%d dB',c)); ylim([-2.5 2.5]); title(sprintf('SNR = %.1f dB',snrdb(m,d)));
end

%% 6. AM and DSB-SC reference (Stage 5): gamma = P_T/(N0 W)
gdb=0:2:40; sa=zeros(size(gdb)); sd=sa;
for k=1:numel(gdb)
   G=10^(gdb(k)/10);
   N0=0.75/G/W;                        % AM, Ac=1, mu=1, PT=0.75
   nI=lpf(sqrt(N0*fs)*randn(N,1),W,f); nQ=lpf(sqrt(N0*fs)*randn(N,1),W,f);
   sa(k)=snrdb(m,sqrt((1+m+nI).^2+nQ.^2));      % envelope detector
   N0d=0.25/G/W;                       % DSB-SC, PT=0.25
   sd(k)=snrdb(m,m+lpf(sqrt(N0d*fs)*randn(N,1),W,f)); % coherent detector
end
figure; plot(gdb,sd,'s',gdb,gdb,'-',gdb,sa,'o',gdb,gdb-4.77,'-',gdb,gdb+10*log10(1.5*beta^2),'k--'); grid on
xlabel('\gamma (dB)'); ylabel('output SNR (dB)'); legend('DSB-SC sim','DSB-SC theory','AM sim','AM theory','FM \beta=5','Location','northwest');

%% 7. Pre-emphasis / de-emphasis (audio-like message, equal rms deviation)
Hpe=1+1j*f/f1; rmsdev=25e3;
ma=real(ifft(fft(real(ifft(fft(randn(N,1)).*(abs(f)<=W))) )./Hpe)); ma=ma/std(ma);
mpe=real(ifft(fft(ma).*Hpe)); sc=std(mpe); mpe=mpe/sc;
cns=[30 25 20 15 12 10]; res=zeros(numel(cns),3);
for k=1:numel(cns)
   y1=addnoise(fmmod(ma,rmsdev,f,fs),cns(k),Bif,f,fs);  d1=lpf(disc(y1,fs)/rmsdev,W,f);
   y2=addnoise(fmmod(mpe,rmsdev,f,fs),cns(k),Bif,f,fs); d2=lpf(disc(y2,fs)/rmsdev,W,f);
```

```
    d2=real(ifft(fft(d2)./Hpe))*sc;                 % de-emphasis
    res(k,:)=[cns(k) snrdb(ma,d1) snrdb(ma,d2)];
end
disp(array2table([res res(:,3)-res(:,2)],'VariableNames',{'CN_dB','noEmph','withEmph','gain'}));
a=W/f1; Iemp=a^3/(3*(a-atan(a))); fprintf('Ideal improvement = %.1f dB\n',10*log10(Iemp));

%% ---------- local functions ----------
function x=fmmod(m,Df,f,fs)
    M=fft(m-mean(m)); ff=f; ff(1)=1; I=M./(1j*2*pi*ff); I(1)=0;
    x=exp(1j*2*pi*Df*real(ifft(I)));                % phase = 2*pi*Df*int(m)
end
function d=disc(x,fs)
    d=angle(x(2:end).*conj(x(1:end-1)))*fs/(2*pi); d=[d; d(end)];
end
function y=lpf(x,B,f)
    X=fft(x); X(abs(f)>B)=0; y=real(ifft(X));
end
function y=addnoise(x,cn_db,B,f,fs)
    s2=10^(-cn_db/10)*fs/B;
    nz=sqrt(s2/2)*(randn(size(x))+1j*randn(size(x)));
    X=fft(x+nz); X(abs(f)>B/2)=0; y=ifft(X);
end
function s=snrdb(ref,est)
    est=est-mean(est); ref=ref-mean(ref); s=10*log10(var(ref)/var(est-ref));
end
```

Appendix B – Key Formulas Used

Quantity	Formula
Modulation index	$\beta = \Delta f / f_m$
Carson bandwidth	$B_T = 2(\Delta f + f_m)$
FM tone SNR	$SNR_o = 1.5\beta^2\gamma$, $\gamma = A_c^2/(2N_0W)$
$\gamma \leftrightarrow$ IF C/N	$\gamma = (C/N)\cdot B_{IF}/W$
AM SNR	$[k_a^2P/(1 + k_a^2P)]\gamma$
DSB-SC SNR	γ
Threshold	$\gamma_{th} \approx 20(\beta + 1)$; $(C/N)_{IF} \approx 10$ dB
Pre-emphasis gain	$I = a^3/\{3(a - \arctan a)\}$, $a = W/f_1$